Sam Boeve

PhD Researcher

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ResearchGate
Google Scholar

Research Interests

Topics	Computational linguistics, Psycholinguistics, Cognitive psychology, Bilingualism, Clinical linguistics, Reading development, Learning
Methods	Large language models, Natural language processing, Mixed effects models

Research Experience

2026-2027	**Computational Psycholinguistics Laboratory (MIT, USA) – Visiting Researcher** 6 month research visit studying predictive sentence processing in dyslexia Supervisor: Prof. dr. Roger Levy
2022-present	**BogaertsLab (UGhent) – PhD Researcher** Predictive reading: context effects across development Promotor: Prof. dr. Louisa Bogaerts, co-promotor: Prof. dr. Hilde van Keer
2021-2022	**DuyckLab (UGhent) – Research Intern** The role of auditory-motor speech processing and higher cognitive abilities in language learning Supervisor: Prof. dr. Eleonore Smalle
2020-2021	**CapLab (UGhent) – Research Assistant** Cognitive and Affective Psychophysiology laboratory: EEG data collection & analysis
2016-2017	**ENdEMIC Lab (UA) – Research Assistant** Environmental Ecology & Applied Microbiology laboratory: SIRM analysis

Education

2020-2022	**Master of Science in Psychology: Theoretical and Experimental Psychology (Neuroscience)** Ghent University (UGhent), Belgium (summa cum laude) Thesis: Can machine learning capture differences in EEG of infants at elevated likelihood and typical likelihood of Autism Promotor: Prof. dr. Herbert Roeyers
2017-2020	**Bachelor of Science in Psychology: Theoretical and Experimental Psychology** UGhent (summa cum laude)
2014-2017	**Bachelor of Science in Bioscience Engineering** University of Antwerp (UA), Belgium (completed 93 credits)
2008-2014	**Secondary school: Sciences – Mathematics** Sint-Gabriëlcollege Boechout

Competencies

Academic Skills	Experimental design, data-analysis, supervision, peer review
Soft Skills	Public speaking, academic writing, science communication
Software	Python (MNE, Transformers, Pytorch), R (mixed models), Julia, PsychoPy, SPSS, MATLAB, Javascript, React Microsoft Office, Latex
Hardware	Electroencephalography (EEG), transcranial direct current stimulation (tDCS), transcranial magnetic stimulation (TMS)
Languages	Dutch (native), English (proficient), French (limited working), German (basic), Spanish (A2)

Grants and Awards

2026	Travel grant for long stay abroad, Fund Scientific Research Flanders (€8000) – MIT (Cambridge, USA)
	Conference grant of Faculty Mobility Fund (~€800) for MMMM conference in Nijmegen
	Travel grant for short stay abroad, Fund Scientific Research Flanders (€1000) – MBZUAI (Abu Dhabi, UAE)
2025	Conference grant of Faculty Mobility Fund (~€800) for TeaP conference in Frankfurt
2024	Poster award (3rd place), Faculty of Psychology and Educational Sciences' Research Day
	Conference grant of Faculty Mobility Fund (~€800) for IASL conference in San Sebastian
2023-2027	Personal PhD fellowship, Fund Scientific Research Flanders - FWO (~€130.000)
	Personal PhD fellowship, Special Research fund, UGhent (~€130.000) [Declined in favour of FWO grant]

Publications

2026

Boeve, S., Figueroa, E. S., & Bogaerts, L. (2026) Prediction Under Pressure: Predictive Sentence Processing in Dyslexia and Second-Language Reading. https://doi.org/10.31234/osf.io/2jz64_v1

Boeve, S., Figueroa, E. S., Badaya, E., Bogaerts, L., & Dirix, N. (2026). Dyslexia Eye Tracking COrpus (DECO): First-and Second Language Book Reading in Adults with Dyslexia. https://doi.org/10.31234/osf.io/m2jf8_v1

2025

Boeve, S., & Bogaerts, L. (2025). A systematic evaluation of Dutch large language models' surprisal estimates in sentence, paragraph and book reading. *Behavior Research Methods,* 57(9), 266. https://doi.org/10.3758/s13428-025-02774-4

Sui, L., Woumans, E., Duyck, W., **Boeve, S.** & Dirix, N. (2025) The Word and Character Frequency Effect in Chinese Natural Reading. *Reading and Writing,* 39(5), 1877-1900. https://doi.org/10.1007/s11145-025-10681-5

Verschooren, S., Vermeylen, L., **Boeve, S.**, & Pourtois, G. (2025). Perceptual Decoupling Underlies Internal Shielding Benefit during Switches between External and Internal Attention: Evidence from Early Sensory Event-related Potential Components. *Journal of Cognitive Neuroscience*, 1-19. https://doi.org/10.1162/jocn_a_02329

2024

Boeve, S., Zhou, H., & Bogaerts, L. (2024). A meta-analysis of 97 studies reveals that statistical learning and language ability are only weakly correlated. *L'Année psychologique*, *124*(3), 283-316. https://doi.org/10.3917/anpsy1.243.0283

Boeve, S., Möttönen, R., & Smalle, E. H. (2024). Specificity of motor contributions to auditory statistical learning. *Journal of Cognition*, *7*(1). https://doi.org/10.5334/joc.351

Talks

2026

Multilingual Minds and Machines Meeting (MMMM, Nijmegen) - Prediction under pressure: How dyslexia and second-language reading shape predictive sentence processing - **Boeve, S.**, Salgado Figueroa, E., Bogaerts, L.

Meeting of Minds for Youth (Breinwijzer vzw, Ghent) – Generatie GenAI – **Boeve, S.**

2025

Workshop Large Language Models for Linguistic Analyses: Applications and Limitations (Cologne) – From Tool to Theory: LLMs in Psycholinguistics – **Boeve, S.**

Conference of Experimental Psychologists (TeaP, Frankfurt) – A Systematic Evaluation of Dutch Large Language Models' Surprisal Estimates – **Boeve, S.**, Bogaerts, L.

2024

Faculty Research Day (Ghent) – Introduction to (gen)AI – **Boeve, S.**

2023

Psycholinguistics in Flanders Conference (PiF, Ghent) – Auditory Statistical Learning and the Role of the Articulatory Motor System – **Boeve, S.**, Möttönen, R., & Smalle, E. H.

Press and Outreach

2025	Article: Language models: A new perspective on language and cognition – *The Inquisitive Mind* – https://in-mind.org/article/language-models-a-new-perspective-on-language-and-cognition
	Article: Taalmodellen: een nieuwe blik op taal en cognitie [Language models: a new perspective on language and cognition] – *The Inquisitive Mind* – https://nl.in-mind.org/article/taalmodellen-een-nieuwe-blik-op-taal-en-cognitie
2024	Podcast: BrainGPT: language in brains and machines – https://www.youtube.com/watch?v=ezdhyu6STNk
	Blogpost: BrainGPT: hersenen upgraden met taalmodellen [BrainGPT: upgrading brains with language models] – *EOS blogs* – https://www.eoswetenschap.eu/technologie/braingpt-hersenen-upgraden-met-taalmodellen
	Article: Taal getransformeerd: AI in onderzoek en onderwijs [Language transformed: AI in research and education] – *Lexima* – https://bit.ly/LexMaga24
2022	Science festival (IBRAIN) on Intelligence - expo booth on implicit learning

Teaching Experience

2024-2025	Teaching assistant: Introduction to Psycholinguistics – Master of Science in Theoretical and Experimental Psychology – Prof. dr. Louisa Bogaerts
	Guest lecture: 'Introduction to (gen)AI' for course Academic Skills – Bachelor of Science in Psychology – Prof. dr. Jan De Houwer
2023-2024	Guest lecture: 'The rise of the machines: large language models and transformers' for course Computational Models in Psycholinguistics – European Master of Science in Clinical Linguistics – Prof. dr. Rob Hartsuiker

Supervision

2023-present	Master thesis	Ella Salgado Figueroa (awarded best master thesis, BAPS 2026) Maria Federica Mottadelli Nickie Ntasinzira Fleur Hooft
	Research project	Sibelle Rasschaert Elise Lyssens

Journal Reviewer

2024-present	Journal of Educational Psychology; Journal of Memory and Cognition; Behavior Research Methods, Scientific Studies of Reading; Journal of Psycholinguistic Research, Reading and Writing

Service

2025	PhD Representative, Departmental Board of Experimental Psychology (Ghent University)
	Member of the organizing committee of the NeuroCog conference (Brussels)
2023	Member of scientific and organizing committee of the Psycholinguistics in Flanders (PiF) meeting (Ghent)
2021	Member of the Education Committee – faculty of Psychology and Educational Sciences (Ghent University)